

KCPC Week 2 Reading List

Advanced Invariants, Heuristics, and Algorithmic Bridges

The Art and Craft of Problem Solving Paul Zeitz (Wiley, 2007)

The definitive guide to Zeitz heuristics: wishful thinking, symmetry, extreme principle, reduction. Chapters on invariants, colouring, and induction are exceptionally clear and directly relevant to both Week 1 and Week 2. The problems and solutions model the kind of deep reasoning we practise.

Problem-Solving Strategies Arthur Engel (Springer, 1998)

A technique-centred encyclopaedia. The chapters on invariants, colouring proofs, and induction are the gold standard for mathematical problem solving. Many of the Week 2 exercises are drawn from or inspired by Engel's examples.

Mathematical Puzzles: A Connoisseur's Collection Peter Winkler (A K Peters, 2004)

Elegant puzzles that often require exactly the invariants, parity arguments, and induction techniques from the workshop. Winkler's solutions are concise and reveal the core insight. Excellent for practising the transition from insight to proof.

Algorithmic Puzzles Anany Levitin and Maria Levitin (Oxford University Press, 2011)

Bridges the gap between mathematical puzzles and algorithms. Organised by technique: state-space search, backtracking, greedy, divide-and-conquer, dynamic programming. Solutions discuss efficiency and implementation, making it ideal for the algorithmic bridges in Week 2.

Guide to Competitive Programming Antti Laaksonen (Springer, 2017)

A compact CP reference. Covers complexity, data structures, graph algorithms, greedy, DP, and more with short C++ implementations. Use this to see how the mathematical ideas from Pair 1 become standard CP algorithms in later pairs. The CSES problem set (cses.fi) is a natural companion.

Introduction to Algorithms: A Creative Approach Udi Manber (Addison-Wesley, 1989)

Organises algorithms around inductive proofs. Shows how a proof about smaller instances directly suggests a recursive or incremental algorithm. Connects Week 1 reasoning with Week 2 bridges to DP, divide-and-conquer, and greedy algorithms.

The Algorithm Design Manual Steven S. Skiena (Springer, 2020)

A practical guide to recognising algorithmic problem types and choosing techniques. The "War Stories" show real modelling decisions. The catalogue of problems helps you map a puzzle to a known algorithmic pattern. Complements the reduction heuristic from Week 2.